

Weeks 13–14: Contract Implementation Summary

Planned Deliverables (from thesis plan)

- Develop SubscriptionManager.rs (state management, payments, verification)
- Implement PayPerView.rs (one-time access, tokens)
- Create PurchaseVerification.rs (ownership transfers, resale)
- Write unit tests (80% coverage) and deploy to local testnet

What Was Actually Built

The original plan envisioned three separate ink! smart contracts. Based on Phase 3 architecture decisions (RMRK 2.0 pallets abandoned, pallet-nfts adopted instead), the implementation was refactored into a **FRAME pallet + ink! contract** architecture:

1. pallet-content-rights (replaces all three planned contracts)

Location: pallets/content-rights/

A unified FRAME pallet that handles all three access models through a single RightsType enum, backed by pallet-nfts for token storage:

Table 1: Planned Contracts vs Pallet Implementation

Planned Contract	Pallet Implementation	Key Extrinsics
SubscriptionManager.rs	RightsType::Subscription	register_content, subscribe, renew_subscription, check_access
PayPerView.rs	RightsType::PayPerView	register_content (with PPV price), pay_per_view, check_access
PurchaseVerification.rs	RightsType::Ownership	register_content, subscribe (permanent), transfer_ownership, transfer_ownership_cross_chain

Relevant commits:

- 5d8ab9a — Add pallet-nfts to runtime for content rights NFT layer
- 8b532b5 — Add pallet-content-rights with full rights management and 23 unit tests
- 2a7962e — Add ownership transfer extrinsics (local and cross-chain)

2. RightsManager ink! 6 Contract

Location: contracts/rights-manager/

An ink! 6 smart contract that provides a higher-level API over the pallet, deployed on pallet-revive (PolkaVM):

- mint_content_rights — Creates a rights NFT for content
- check_access — Verifies access via custom precompile

Relevant commits:

- f4bc50c — Scaffold RightsManager ink! 6 contract
- 7bc9651 — Add custom pallet-revive precompile for content rights read access

3. Unit Tests

23 unit tests in `pallets/content-rights/src/tests.rs` covering:

- Content registration (all three types)
- Subscription creation and renewal
- Pay-per-view access and counter tracking
- Ownership transfers (local and cross-chain)
- Expiry and access checks
- Error paths (unauthorized, not found, expired)

Relevant commit: 8b532b5

4. Local Testnet Deployment

Deployed and verified on Zombienet local testnet:

- 2-chain topology (Rococo relay + Content Rights parachain 100)
- Verified via `scripts/verify-runtime-has-contracts.sh`

Deviations from Plan

Table 2: Deviations from Plan		
Planned	Actual	Rationale
Three separate ink! contracts	One FRAME pallet + one ink! contract	FRAME pallets are more efficient for core state management; ink! provides the external API layer
ink! on pallet-contracts (WASM)	ink! 6 on pallet-revive (PolkaVM)	pallet-revive is the current Polkadot SDK direction; pallet-contracts is legacy
RMRK 2.0 for NFT nesting	pallet-nfts with attribute-based semantics	RMRK pallets abandoned and incompatible with current polkadot-sdk